Black-Box 5-State a'_x Estimator — Consolidated Current Version

Status. Consolidates `5state_est_aprime.tex` + `5state_aprime_selfmod.tex` (F_e/H algebra, reused verbatim, Sec. 3–5) with `aprime_blackbox_regression.tex` (Isserlis unbiasedness of a_{xm}, Sec. 2) and three NEW pieces from this session: the $Q(a'_x)$ formula (Sec. 7, empirically checked), the δx_r-vs-$\delta\hat{x}$ signal reconciliation (Sec. 6, leading explanation for a still-open sign-flip bug), and a concrete near-wall reversion fix (Sec. 8). **Not yet implemented in code** – this is the design document that precedes implementation.

1. Setting: the black-box premise

	Known-wall 4-state (`4state_del_hd_ar1`)	**This document (black-box)**
a_x predict target	$a_{x_{det}}[k] = a_{nom}/c(\bar{h}_d[k])$	none — integrated state, no external target
a'_x	$-a_x K_h/R$, from the real $c(\bar{h})$	*estimated*, 5th state
K_h, K'_h	`calc_correction_functions`	forbidden anywhere, incl. t=0 seeding
Spatial memory	—	none; purely time-recursive (no $\bar{h}$-indexed map)

Wall *location* is known ($\bar{h}[k]$ measurable from $p_m[k]$ and the known wall plane); only the *function* $c(\bar{h})$ is unknown. Excitation requirement (per design discussion): the estimator must work **both** during commanded height motion *and* during a static hold.

2. Why a_{xm} is a legitimate, unbiased measurement (imported result)

$a_{xm}[k] = a_x[k-d]+n_a[k]$ is built by squaring and EWMA-averaging the residual δx_r (`pm_to_axm_derivation.tex`) so a natural worry is circularity: is a_{xm}'s own noise n_a correlated with the very state (a_x, or its fluctuation) we want to measure? `aprime_blackbox_regression.tex` resolved this by Isserlis' theorem (zero-mean jointly Gaussian vectors have zero odd-order moments): every term contributing to a_{xm} is a *quadratic* functional of δx_r, so its covariance with δx_r itself (degree 3, odd) vanishes exactly, regardless of a_{cov}, the LP bandwidth, or δx_r's autocorrelation structure. This is why a_{xm} can be treated below as a clean linear measurement of $a_x[k-d]$ with additive noise n_a, not a self-referential trap.

3. Parameter model (imported from 5state_aprime_selfmod, verbatim)

Local decomposition (derivation device only — **not** evaluated at runtime):

$$a_x[k] = a_{\rm det}[k] + a_{x_{ram}}[k], \qquad a_{x_{ram}}[k] = a'_x\, x_{ram}[k] = -\, a'_x\, \delta x[k]$$

Closing the loop ($\delta x_3 = \delta x[k]$ the current tracking error, ε the lumped closed-loop noise defined in Sec. 4):

$$\delta x[k+1] = \lambda_c\, \delta x_3[k] - \varepsilon[k] \;\Rightarrow\; \Delta\delta x[k] = -(1-\lambda_c)\, \delta x_3[k] - \varepsilon[k]$$

$$\boxed{\begin{cases} a_x[k+1] = a_x[k] + a'_x\big[\Delta h_d[k] + (1-\lambda_c)\,\delta x_3[k]\big] + w_4[k] \\ a'_x[k+1] = a'_x[k] + w_{a'}[k] \end{cases}}$$

$\Delta h_d[k] = h_d[k{+}1] - h_d[k]$ (known, from the commanded trajectory) supplies observability **during motion**; $(1-\lambda_c)\,\delta x_3[k]$ (thermal self-dither, zero-mean, $\sim$tens of nm) supplies observability **on a hold**. Neither requires $c(\bar{h})$.

4. State equation (5-state)

$$\begin{aligned}
&\delta x_1[k+1] = \delta x_2[k], \qquad \delta x_2[k+1] = \delta x_3[k], \\
&\delta x_3[k+1] = \lambda_c\,\delta x_3[k] - F_{dx}[k]\,e_{a_x}[k] + dF^h_{dx}[k]\,e_{a'}[k] - \varepsilon[k], \\
&\ a_x[k+1] = a_x[k] + a'_x\big[\Delta h_d[k] + (1-\lambda_c)\,\delta x_3[k]\big] + w_4[k], \\
&\ a'_x[k+1] = a'_x[k] + w_{a'}[k].
\end{aligned}$$

$$F_{dx}[k] = f_{dx}[k] + (1-\lambda_c)\sum_{i=1}^{d} f_{dx}[k-i], \qquad dF^h_{dx}[k] = (1-\lambda_c)\sum_{i=1}^{d}\big(h_d[k]-h_d[k-i]\big)\,f_{dx}[k-i]$$

$e_{a_x} = a_x - \hat{a}_x$, $e_{a'} = a'_x - \hat{a}'_x$ (level and slope estimation errors).

5. Output equation, H, closed-loop estimator, $\mathbf{F}_e$ (imported verbatim)

$$a_x[k-d] = a_x[k] - a'_x\big[\Delta H_d[k] - \big(\delta x_3[k] - \delta x_1[k]\big)\big], \qquad \Delta H_d[k] = h_d[k] - h_d[k-d]$$

$$y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]},$$

$$y_2[k] = a_{xm}[k] = a_x[k-d] + n_a[k] \approx a_x[k] - a'_x\big[\Delta H_d[k] - \big(\delta x_3[k] - \delta x_1[k]\big)\big] + \underbrace{n_a[k]}_{r_2[k]}.$$

The span $a'_x(\delta x_3 - \delta x_1)$ is carried entirely by **H** (not left in r_2) — this is the fix for the "common thermal noise double-count" that self-mod's Caveat 2 documents (keeping the span in *both* **H** and R_{22} self-defeats the very observability the self-dither term is supposed to provide).

$$\boxed{\mathbf{H}[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -\hat{a}'_x & 0 & \hat{a}'_x & 1 & -\Delta H_d[k] + (\delta\hat{x}_3 - \delta\hat{x}_1) \end{bmatrix}}$$

($\mathbf{H}(2,1), \mathbf{H}(2,3) = \mp\hat{a}'_x$ are $O(\hat{a}'_x)$, negligible; the load-bearing term is $\mathbf{H}(2,5)$.)

Error dynamics ($G_4[k] := \Delta h_d[k] + (1-\lambda_c)\delta\hat{x}_3[k]$, $G_H[k] := \Delta H_d[k] - (\delta\hat{x}_3 - \delta\hat{x}_1)$):

$$\boxed{\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] & dF^h_{dx}[k] \\ 0 & 0 & 0 & 1 & G_4[k] \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}}$$

$\mathbf{F}_e(4,5) = G_4[k]$ is the direct consequence of a_x's own predict equation using $\hat{a}'_x$ (Sec. 3) — *not* an assumption bolted on. $\mathbf{F}_e(3,4) = -F_{dx}$ is unchanged from the 4-state (control law uses $\hat{a}_x$ to compute force). a'_x affects δx_3 only *indirectly*, through a_x's error (two-hop propagation); there is no direct $\mathbf{F}_e(3,5)$ unless $\hat{a}'_x$ is also fed forward into the control law directly (a separate design choice, not made here).

6. Reconciliation: raw δx_r vs. filtered $\delta\hat{x}_3 - \delta\hat{x}_1$ — leading explanation for the sign-flip

This session's external, standalone regression test (`verify_aprime_blackbox_v1_ensemble.m`) computed

$$\hat{a}'_{\mathrm{ext}} = -\frac{\mathrm{Cov}(a_{xm}, \delta x_r)}{\mathrm{Var}(\delta x_r)}$$

using the **raw** residual δx_r (the a_{xm} channel's own IIR intermediate, *before* any state filtering) as the Δh proxy. Against 60 independent seeds (static hold, $\bar{h} \approx 1.78$), this gave a statistically significant **wrong-sign** result (t-stat -2.9 to -7.6), even though: (i) the pure regression algebra was verified unbiased on synthetic data built to the same model (`verify_aprime_blackbox_synthetic.m`, both i.i.d. and autocorrelated δx_r), (ii) a d-step delay misalignment does not by itself flip the sign (`verify_aprime_blackbox_synthetic_delay.m`), and (iii) an oracle indexing fix (pre/post-integration alignment) changed nothing.

The present derivation (Sec. 5, imported from `5state_aprime_selfmod`) uses $\delta\hat{x}_3 - \delta\hat{x}_1$ — the **EKF's own filtered state estimates** of $\delta x_3, \delta x_1$ — not the raw δx_r. These are related but **not** the same signal: $\delta\hat{x}_3, \delta\hat{x}_1$ are shaped by the *closed-loop feedback* (the control law computes $f_{dx}[k]$ directly from $\delta x_m[k]$, which the raw δx_r also feeds into, but $\delta\hat{x}$ is the KF's own posterior after fusing y_1 and y_2, weighted by the current $\mathbf{P}$). **Leading (unconfirmed) hypothesis**: the external test's raw-δx_r regressor absorbs contamination from this closed-loop feedback path (the control law's use of $\delta x_m[k]$ to compute the very force that determines the next position) that the properly-derived $\mathbf{H}(2,5)$ — built from the filtered $\delta\hat{x}$, and appearing *inside* a jointly-consistent $\mathbf{F}_e/\mathbf{H}$ pair — does not suffer, because the KF's own error-dynamics bookkeeping (Sec. 5) already accounts for this coupling structurally rather than assuming it away via an external statistical identity.

Confirmed (`verify_5state_aprime_selfmod_signflip.m`, same static-hold scenario, real controller with `use_selfmod=true`): at 60 seeds, $\hat{a}'_x$ (read directly from `diag.delta_a_hat`) already showed $|t\text{-stat}| < 1.51$ on all three axes versus the external test's -2.9 to -7.6; the y, z axes matched the oracle's sign, while x's point estimate was nominally opposite but statistically indistinguishable from zero (SE exceeded |mean|). A re-run at 200 seeds resolved this: all three axes now match the oracle's sign (x, y, z: $t = -0.62,\ 0.01,\ -1.24$), confirming the x-axis discrepancy was sampling noise, not a residual sign problem. This confirms the raw-vs-filtered-signal distinction (this section's hypothesis) was the dominant explanation for the external test's wrong sign.

7. $Q(a'_x) = q_5$ — honest process-noise design (NEW this session)

`5state_aprime_selfmod.tex` leaves $q_5 = \mathrm{Var}(w_{a'})$ labelled only as "the estimator bandwidth on the slope," with no formula. Decompose:

$$\boxed{q_5[k] = \sigma^2_{a''} \cdot \left(\Delta h_d[k]^2 + \sigma^2_{\delta h}[k]\right), \qquad \sigma^2_{\delta h}[k] = 4k_BT\,\hat{a}_x[k]}$$

Rationale. a'_x changes because the (unknown) curvature $a''_x = da'_x/d\bar{h}$ is nonzero; moving by Δh changes a'_x by $\approx a''_x \cdot \Delta h$ (chain rule), so $\mathrm{Var}(\Delta a'_x) \approx a''^2_x \cdot \mathrm{Var}(\Delta h)$. $\Delta h_d[k]$ (commanded) is exactly known; $\sigma^2_{\delta h}[k]$ (thermal) reuses the already-validated $4k_BT\,a_x$ formula with the **current level estimate** $\hat{a}_x[k]$ (honest, no wall peek). Only $\sigma^2_{a''}$ — one scalar, the curvature magnitude prior — remains free.

Dimensional anchor for $\sigma^2_{a''}$

R (probe radius) and $a_{nom} = \Delta t/\gamma_N$ are known probe/fluid constants, **not** wall properties. Since $c(\bar{h})$ is defined as a function of $\bar{h} = h/R$, R is the wall effect's own natural length scale:

$$\sigma_{a'} \sim a_{nom}/R \quad \text{(slope scale)}, \qquad \sigma_{a''} \sim \sigma_{a'}/R = a_{nom}/R^2 \quad \text{(curvature scale)}$$

$$\boxed{\sigma^2_{a''} \sim (a_{nom}/R^2)^2}$$

This is an order-of-magnitude engineering prior, not an exact derivation — but it is *derived from known constants*, not picked from nothing.

Empirical check (verify_aprime_Qprime_assumptions.m, run before this file was written)

	near-wall ($\bar{h} \approx 1.05$–1.2)	$\bar{h} \approx 1.78$	far-field ($\bar{h} > 5$)
$\lvert a'_{true}\rvert/(a_{nom}/R)$	0.6–0.9	0.30	≤ 0.04
$\lvert a''_{true}\rvert/(a_{nom}/R^2)$	2.2–3.8	0.31	≤ 0.02
$\mathrm{Var}(\Delta a'_{true})$ vs. predicted (real osc trajectory)	ratio 0.65	—	ratio $\sim 10^{-4}$

Near-wall (where a'_x matters most): the anchor is the correct order of magnitude (ratios 0.6–3.8×, and the $\mathrm{Var}(\Delta a')$ check lands within 35%). **Far-field: the fixed constant over-covers by $\sim 10^4\times$** — the already-known, architecturally unavoidable single-constant trade-off (Sec. 8 of `5state_aprime_selfmod.tex`'s antecedent discussion), now quantified with real numbers. Over-covering far-field is low-cost: $a'_x \to 0$ there and tracking does not depend on it.

8. Caveat 5 fix proposal: state-computable near-wall reversion (NEW this session)

`5state_aprime_selfmod.tex` Caveat 5: $\mathbf{F}_e(4,4) = 1$ is a free integrator (no reversion target, since there is no known a_{det}); when the near-wall gate turns a_{xm} off, a_x has neither reversion nor measurement and drifts as an undamped random walk. It explicitly flags the fix as required-but-undone: *"a state-computable reversion ($a_{x_{ram}} = -a'_x \delta x_3$) ... is required before closing the loop."*

Proposal

Use the **already-known local-linear relationship** (Sec. 3) as a self-consistency check, not an external target. Define a slow reference level $\bar{a}_x[k]$ (its own, slower EWMA of $\hat{a}_x$ — purely internal, no wall model), and add a weak correction pulling $\hat{a}_x$ toward the value **implied by the current state itself**:

$$\boxed{\hat{a}_x[k+1] = \hat{a}_x[k] + \hat{a}'_x[k]\big[\Delta h_d[k] + (1-\lambda_c)\delta\hat{x}_3[k]\big] + \kappa\Big(\underbrace{\bar{a}_x[k] - \hat{a}'_x[k]\,\delta\hat{x}_3[k]}_{\text{state-implied } a_{\text{det}} \text{ estimate}} - \hat{a}_x[k]\Big)}$$

κ small (weak pull, e.g. $\kappa \sim (1-\lambda_c)/N$ for some large averaging horizon N — a design choice, not yet tuned). Physical bound as a secondary safety net (fully honest, R/a_{nom} are probe constants): $0 < a_x \le a_{nom}$ always (the wall only adds drag, never removes it below the free-space value); a clip or soft-barrier reparametrization on $\hat{a}_x$ prevents runaway growth even if κ is mistuned. **Not yet derived to the same rigor as Sec. 3–5; flagged here as the next piece of derivation work, not a finished result.**

9. Consolidated verification status

Test	What it checked	Result
V0 (free-space null)	Isserlis unbiasedness on real simulated signal	**PASS**
V1 (1 Hz osc, single traj.)	external regression vs. oracle	bug found (wrong regressor axis) + locality-vs-SNR tension found
V1-hold (static, single traj.)	same, locality removed	weak corr., large bias; ambiguous (noise vs. real)
V1-ensemble (60 seeds, batch)	same, high power	significant wrong-sign ($t = -2.9$ to -7.6); resolved by the two rows below
Synthetic (pure algebra)	regression formula + Isserlis, ρ=0, 0.85	**PASS** (rules out formula bug)
Synthetic+delay	d-step misalignment alone	does not flip sign (rules out this cause alone)
Oracle pre/post-integration fix	indexing bug	no numerical effect (rules out this cause)
$Q(a')$ magnitude (Sec. 7)	dimensional anchor vs. oracle	**near-wall PASS** (order of magnitude); far-field over-covers (expected)
5-state `use_selfmod` (60 seeds, real controller)	same scenario, real `diag.delta_a_hat`	sign correct on y, z; x nominally wrong but $\|t\| < 1$ (noise, SE> |mean|)
5-state `use_selfmod` (200 seeds, real controller)	same scenario, higher power	**sign CONFIRMED correct on all 3 axes** ($t = -0.62$, 0.01, -1.24)

Resolved: the wrong-sign result was caused by the raw-δx_r-vs-filtered-signal distinction (Sec. 6). The properly-derived 5-state EKF (Sec. 3–5,7), read directly from the real controller (`use_selfmod=true`), recovers the correct sign on all three axes at 200 seeds. **Not yet done**: accuracy characterization beyond sign (L1-style κ sweep, hold vs. motion, near vs. far field); the Sec. 8 near-wall reversion proposal remains undone (this test's scenario does not exercise the gate-off condition); Q33/Q44/R22 remain wall-peeking (Sec. 7 note).